

Transfer and Extension of n -Overarching Axioms Across Structures

Abstract

This paper formalizes the conditions under which n -overarching axioms are preserved when extending mathematical structures from subsets to supersets. We introduce a rigorous definition for the *axiomatic domain extension* of functions, requiring the preservation of semantic validity rather than mere syntactic arity. Furthermore, we propose a fourth law, the Law of Uniform Sort Scaling, and establish a taxonomy of failure modes that prevent successful axiomatic extension, thereby addressing the gaps left by the existing laws of overarching axioms.

1 Introduction

In the study of formal systems, an n -overarching axiom is defined as a statement that ranges over n distinct variables or elements within a domain. In the existing framework, Theorem 1 (Resolution Scaling) handles collapsing ∞ -overarching axioms down via containerization (higher-order packaging reducing arity). However, it lacks a companion theorem for axioms that keep the same arity syntactically but change density semantically when the sorts they range over are embedded into a strictly higher-dimensional superstructure. This paper addresses that gap by shifting the focus from the trivial preservation of syntactic count to the preservation of *validity*, treating n -overarching-ness as a side-condition rather than the conclusion.

2 Preservation of Validity Across Subsets

Theorem 1 (Subset Dependency). *An n -overarching axiom of a subset $S' \subseteq S$ is also n -overarching on S if and only if it only depends on axioms in both S' and S .*

Proof. Let ϕ be an n -overarching axiom on S' . By definition, ϕ binds n distinct variables over S' . For ϕ to remain valid and n -overarching on S , it must maintain non-vacuous applicability on S . If ϕ depends only on axioms that hold in both S' and S , then the truth conditions of ϕ are preserved when the domain of the bound variables is expanded from S' to S .

Conversely, if ϕ is n -overarching on S and its restriction to S' holds, it must not rely on axioms exclusive to $S \setminus S'$, as those would not be guaranteed to hold when restricted to the subset. Thus, the axiom depends only on axioms present in both. $\square$

3 Failure Modes and Dimensional Promotion

When an axiom is transferred between sets, its syntactic arity n might be trivially preserved, but its semantic validity might fail. We identify four primary failure modes that prevent an axiom from being non-vacuously true on a superstructure G :

1. **Closure:** The superstructure lacks closure under the operations required by the axiom.

2. **Totality:** The extended operations are not defined for all elements in G .
3. **Injectivity:** The extension causes a loss of one-to-one correspondence, rendering certain axioms vacuous or false.
4. **Sort-Density Mismatch:** The embedding scales ambient sorts at different rates, disrupting the relative dimensional density required by the axiom.

To address the sort-density mismatch, we propose a fourth law to accompany the existing three Laws of n -overarching axioms.

Law 1 (Uniform Sort Scaling). *n -overarching status is preserved under an embedding $T : S \hookrightarrow S'$ only if the relative dimensional density of every sort bound by the axiom is preserved under the embedding T . If a bound point shares the same domain sort and scales uniformly under extension, no auxiliary sort is involved and transfer holds as expected.*

4 Axiomatic Domain Extension

We now formalize the extension of functions and their associated axioms. The goal is to preserve the substantive content and truth of the axioms, rather than merely their syntactic arity.

Definition 1 (Axiomatic Domain Extension). *Let $f : H \rightarrow K$ be a function, and let J be an axiom group relating to f . A function $h : G \rightarrow K'$ is an axiomatic domain extension of f with axiom group J if:*

1. $H \subseteq G$,
2. $K \subseteq K'$,
3. $h|_H = f$ (the restriction of h to H equals f),
4. K' has the same definition of equality as K (i.e., the $=$ sign means the same thing in both, though K' may contain additional equivalence relations or isomorphisms),
5. All n -overarching axioms that relate to f in J are also overarching in G upon replacing f with h .

If no such function h exists that satisfies these conditions, then f physically cannot be axiomatically domain extended. This definition provides a rigorous pass/fail test for validity preservation, while the taxonomy of failure modes (Section 3) provides the diagnostic criteria for why an extension fails.

5 Examples in Real and Complex Spaces

Example 1 (Linearity and Sort-Density Mismatch). *Consider the 2-overarching axiom $x = ty$ for some $t \in \mathbb{R}$ in the real space $\mathbb{R}$. This axiom depends on the linearity of the real space. When embedded into the complex space $\mathbb{C}$, this overarching axiom fails. The failure occurs because the overarching axiom depends on the linearity of the real space, which experiences a sort-density mismatch when promoted to the strictly higher-dimensional complex superstructure. Specifically, an embedding that scales two different ambient sorts at different rates, such as $T : \mathbb{R}^2 \hookrightarrow \mathbb{R}^3 = \mathbb{R} \times \mathbb{C}$, disrupts the relative dimensional density.*

Example 2 (Exponential and Logarithmic Identities). *Consider the axiom $x = e^{\ln(x)}$ on the real space $\mathbb{R}$. When extended to the complex space $\mathbb{C}$, the identity $e^{\ln(z)} = z$ transfers cleanly. There is only one bound variable z , so it moves uniformly from $\mathbb{R}$ to $\mathbb{C}$. By the Law of Uniform Sort Scaling, no auxiliary sort is involved, and the transfer holds as expected, confirming the theorem.*

However, the converse identity $\ln(e^x) = x$ fails in $\mathbb{C}$. The complex logarithm is multivalued, meaning $\ln(e^z)$ does not necessarily equal z for all $z \in \mathbb{C}$. This failure is correctly predicted by the totality and injectivity failure modes identified in Section 3, as the extension fails to preserve non-vacuous truth across the domain.

6 Conclusion

The framework established here provides a rigorous criterion for distinguishing cosmetic relabeling from genuine axiom changes during structural extension. By defining axiomatic domain extension with strict codomain superset and equality constraints, and by classifying the failure modes, we provide a comprehensive apparatus for evaluating the preservation of validity across domains.